US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250282034
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	ZHAO; Ding

ELECTRIC SCREW TIGHTENING MACHINE AND ANGLE RATCHET WRENCH

Abstract

An electric screw tightening machine includes: a motor including a stator and a rotor that is rotatable with respect to the stator and has a rotor axis extending in a front-rear direction; a motor housing that accommodates the motor; a speed reduction mechanism that decelerates rotation of the motor; a socket to which rotation of the motor is always transmitted via the speed reduction mechanism during rotation of the motor and that has a socket axis intersecting the rotor axis; and a one-way clutch mechanism that is disposed at least partially around the socket, and is configured to allow the socket to rotate due to the rotation of the motor only in one direction during the rotation of the motor and transmit rotation input to the motor housing to the socket only in one direction during stop of the motor.

Inventors: **ZHAO; Ding (Anjo-shi, JP)**

Applicant: **MAKITA CORPORATION** (Anjo-shi, JP)

Family ID: **1000008448700**

Assignee: **MAKITA CORPORATION (Anjo-shi, JP)**

Appl. No.: **19/051313**

Filed: **February 12, 2025**

Foreign Application Priority Data

JP	2024-033208	Mar. 05, 2024

Publication Classification

Int. Cl.: **B25B21/00** (20060101); **B25B13/46** (20060101)

U.S. Cl.:

CPC **B25B21/004** (20130101); **B25B13/463** (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] The present application claims priority to and incorporates by reference the entire contents of Japanese Patent Application No. 2024-033208 filed in Japan on Mar. 5, 2024.

TECHNICAL FIELD

[0002] The techniques disclosed in the present specification relate to an electric screw tightening machine and an angle ratchet wrench.

BACKGROUND ART

[0003] In the technical field related to an electric screw tightening machine, an electric ratchet wrench as disclosed in WO 2019/026699 A is known.

[0004] As described in WO 2019/026699 A, the electric ratchet wrench once converts the rotation of the motor into a reciprocating motion and converts the reciprocating motion into a rotational motion of the socket. In the reciprocating motion caused by the rotation of the motor, for example, the socket rotates in the forward rotation direction during the forward motion, and the socket does not rotate in the forward rotation direction during the backward motion. Therefore, the socket contributes to the fastening work during the forward motion, but the socket does not contribute to the fastening work during the backward motion. Therefore, transmission efficiency of the rotational force from the motor to the socket is low (for example, about 50%). When the transmission efficiency of the rotational force is low, the work efficiency of the electric ratchet wrench is low.

SUMMARY

[0005] One non-limiting object of the present teachings is to suppress a decrease in work efficiency.

[0006] In one non-limiting aspect of the present teachings, an electric screw tightening machine includes: a motor including a stator and a rotor that is rotatable with respect to the stator and has a rotor axis extending in a front-rear direction; a motor housing that accommodates the motor; a speed reduction mechanism that decelerates rotation of the motor; a socket to which rotation of the motor is always transmitted via the speed reduction mechanism during rotation of the motor and that has a socket axis intersecting the rotor axis; and a one-way clutch mechanism that is disposed at least partially around the socket, and is configured to allow the socket to rotate due to the rotation of the motor only in one direction during the rotation of the motor and transmit rotation input to the motor housing to the socket only in one direction during stop of the motor.

[0007] According to the techniques disclosed in the present specification, a decrease in work efficiency is suppressed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is a side view illustrating an electric screw tightening machine according to an embodiment;

[0009] FIG. **2** is a perspective view from the front illustrating part of the electric screw tightening machine according to the embodiment;

[0010] FIG. **3** is a longitudinal sectional view illustrating the electric screw tightening machine according to the embodiment;

[0011] FIG. **4** is a block diagram illustrating the electric screw tightening machine according to the

embodiment;
[0012] FIG. **5** is a perspective view from the front illustrating an output unit according to the embodiment;
[0013] FIG. **6** is a longitudinal sectional view illustrating the output unit according to the embodiment;
[0014] FIG. **7** is a transverse cross-sectional view illustrating the output unit according to the embodiment;
[0015] FIG. **8** is an exploded perspective view from above illustrating the output unit according to the embodiment;
[0016] FIG. **9** is an exploded perspective view from below illustrating the output unit according to the embodiment;
[0017] FIG. **10** is a perspective view in which a main part of the output unit according to the embodiment is extracted;
[0018] FIG. **11** is a view of a ratchet head according to the embodiment when viewed from below; and
[0019] FIG. **12** is a side view illustrating the electric screw tightening machine according to the embodiment.
DETAILED DESCRIPTION
[0020] In one or more embodiments, an electric screw tightening machine includes: a motor including a stator and a rotor that is rotatable with respect to the stator and has a rotor axis extending in a front-rear direction; a motor housing that accommodates the motor; a speed reduction mechanism that decelerates rotation of the motor; a socket to which rotation of the motor is always transmitted via the speed reduction mechanism during rotation of the motor and that has a socket axis intersecting the rotor axis; and a one-way clutch mechanism that is disposed at least partially around the socket, and is configured to allow the socket to rotate due to the rotation of the motor only in one direction during the rotation of the motor and transmit rotation input to the motor housing to the socket only in one direction during stop of the motor.
[0021] In the above configuration, when the screw is fastened, the socket is rotated by the rotational force of the motor. Since the motor rotates at a high speed, the socket also rotates at a high speed. Since the socket rotates at a high speed, the screw is tightened in a short time. During stop of the motor, the electric screw tightening machine can function as a manual ratchet wrench. Therefore, after tightening the screw by the rotational force of the motor, the user of the electric screw tightening machine can use the electric screw tightening machine with the motor stopped as a manual ratchet wrench to additionally tighten the screw. That is, after tightening the screw at a high speed to some extent by the rotational force of the motor, the user can additionally tighten the screw by using the electric screw tightening machine with the motor stopped as a manual ratchet wrench. Therefore, a decrease in work efficiency is suppressed.
[0022] In one or more embodiments, during rotation of the motor, the socket rotates at 800 rpm or more.
[0023] In the above configuration, since the socket rotates at a high speed, the tightening work is completed in a short time.
[0024] In one or more embodiments, the electric screw tightening machine functions as a manual ratchet wrench during the stop of the motor.
[0025] In the above configuration, after tightening the screw at a high speed by the rotational force of the motor, the user can additionally tighten the screw with the electric screw tightening machine with the motor stopped.
[0026] In one or more embodiments, the speed reduction mechanism includes: a first bevel gear that rotates about the rotor axis; and a second bevel gear that is rotationally fixed to the socket, meshes with the first bevel gear, and rotates about the socket axis.
[0027] In the above configuration, since the rotor axis and the socket axis are orthogonal to each

other, the electric screw tightening machine can be used as a manual ratchet wrench. The rotational force of the motor is transmitted to the socket via the first bevel gear and the second bevel gear.
[0028] In one or more embodiments, the electric screw tightening machine includes an intermediate shaft configured to couple the planetary gear mechanism and the first bevel gear of the speed reduction mechanism. The electric screw tightening machine is configured such that rotation of the speed reduction mechanism is transmitted to the first bevel gear via the intermediate shaft, and the rotation of the speed reduction mechanism is transmitted to the first bevel gear without via the intermediate shaft.
[0029] In the above configuration, the dimension of the electric screw tightening machine in the front-rear direction can be arbitrarily adjusted.
[0030] In one or more embodiments, the one-way clutch mechanism allows the socket to rotate in the free direction and blocks the socket from rotating in the lock direction. The one-way clutch mechanism is configured to switch between the free direction and the lock direction. The electric screw tightening machine includes an operation member that is operated to switch between a free direction and a lock direction.
[0031] In the above configuration, the user can perform, for example, both the screw tightening work and the screw loosening work using the electric screw tightening machine.
[0032] In one or more embodiments, an electric screw tightening machine includes: a sensor that detects an operation state of an operation member; and a controller to which a detection signal of the sensor is input. The controller controls the rotation direction of the motor such that the motor rotates the socket in the forward rotation direction when determining that the forward rotation direction of the socket is the free direction of the socket based on the detection signal of the sensor, and the motor rotates the socket in the reverse rotation direction when determining that the reverse rotation direction of the socket is the free direction of the socket.
[0033] In the above configuration, the controller can control the rotation direction of the motor so that the rotational force of the motor is not transmitted in the lock direction of the socket but is transmitted in the free direction. In addition, after tightening the screw by the rotational force of the motor, the user can additionally tighten the screw with an electric screw tightening machine that functions as a manual ratchet wrench without operating the operation member.
[0034] In one or more embodiments, the one-way clutch mechanism includes: a cam plate that is disposed at least partially around the socket and rotates about a socket axis by an operation of the operation member; a first ratchet pawl that is movably supported by an upper face of the cam plate and meshes with a gear of the socket; a second ratchet pawl that is movably supported by an upper face of the cam plate and meshes with a gear of the socket; a first spring that generates an elastic force so as to press the first ratchet pawl against a cam projection provided on the upper face of the cam plate; and a second spring that generates an elastic force so as to press the second ratchet pawl against the cam projection. When the cam plate rotates in the forward rotation direction, the first ratchet pawl is away from the socket, the reverse rotation direction is the free direction of the socket, and the forward rotation direction is the lock direction of the socket. When the cam plate rotates in the reverse rotation direction, the second ratchet pawl is away from the socket, the forward rotation direction is the free direction of the socket, and the reverse rotation direction is the lock direction of the socket.
[0035] In the above configuration, the free direction and the lock direction of the socket are switched only by operating the operation member.
[0036] In one or more embodiments, the electric screw tightening machine includes a positioning ball that is to be disposed in a first recess provided in the cam plate when the cam plate is rotated in the forward rotation direction and that is to be disposed in a second recess provided in the cam plate when the cam plate is rotated in the reverse rotation direction.
[0037] In the above configuration, the cam plate is positioned by the positioning ball.
[0038] Hereinafter, embodiments will be described with reference to the drawings. In the

embodiment, the positional relationship of each part will be described using terms of left, right, front, rear, up, and down. These terms indicate the relative position or the direction with respect to the center of an electric screw tightening machine **1**. The electric screw tightening machine **1** includes a motor **2** as a power source. In the embodiment, the rotor axis AX indicating the rotation axis of the motor **2** extends in the front-rear direction.

Electric Screw Tightening Machine

[0039] FIG. **1** is a side view illustrating the electric screw tightening machine **1** according to the embodiment. FIG. **2** is a perspective view from the front the illustrating part of the electric screw tightening machine **1** according to the embodiment. FIG. **3** is a longitudinal sectional view illustrating the electric screw tightening machine **1** according to the embodiment. FIG. **4** is a block diagram illustrating the electric screw tightening machine **1** according to the embodiment.

[0040] The electric screw tightening machine **1** is an angle ratchet wrench. The electric screw tightening machine **1** includes the motor **2**, an output unit **3**, a power transmission mechanism **4**, a motor case **5**, a motor bracket **6**, a gear case **7**, a lock nut **8**, an extension case **9**, a ratchet head **10**, a motor housing **11**, an operation member **12**, a battery mounting portion **13**, a sensor **27**, and a controller **50**.

[0041] The motor **2** is a power source of the electric screw tightening machine **1**. The motor **2** is an inner rotor type DC brushless motor. The motor **2** extends in the front-rear direction. The rotor axis AX extends in the front-rear direction. The motor **2** generates a rotational force about the rotor axis AX. The motor **2** is accommodated in the motor case **5**. The motor bracket **6** is disposed between the motor case **5** and the gear case **7**. The motor bracket **6** is disposed inside the motor case **5**. The motor bracket **6** is disposed so as to cover the opening at the rear end portion of the gear case **7**.

[0042] The motor **2** includes a stator **2**A, a rotor **2**B having the rotor axis AX that is rotatable with respect to the stator **2**A and extends in the front-rear direction, and a rotor shaft **2**C fixed to the rotor **2**B. The stator **2**A includes a stator core, an insulator fixed to the stator core, and coils respectively wound around teeth of the stator core via the insulator. The rotor **2**B is disposed radially inside the stator **2**A. The rotor **2**B is rotatable at a position radially inside the stator **2**A. The rotor **2**B includes a rotor core and a magnet fixed to the rotor core. The rotor **2**B is disposed around the rotor shaft **2**C. The rotor shaft **2**C is long in the front-rear direction. The central axis of the rotor shaft **2**C coincides with the rotor axis AX.

[0043] A sensor substrate **2**D on which three Hall ICs (magnetic sensors) are mounted is fixed to a rear portion of the stator **2**A. The Hall ICs detect the rotation of the rotor **2**B by detecting the magnetic force of the magnet of the rotor **2**B. The detection signals of the Hall ICs are transmitted to the controller **50**. The controller **50** controls the rotation of the motor **2** based on the detection signals from the Hall ICs.

[0044] The output unit **3** includes a socket **14** having a socket axis CX, and a one-way clutch mechanism **15** that allows the socket **14** to rotate in the free direction and blocks the socket from rotating in the lock direction. The socket axis CX intersects with the rotor axis AX. The socket axis CX is orthogonal to the rotor axis AX. The socket axis CX extends in the vertical direction. The socket **14** is rotatable about the socket axis CX. The socket **14** and the one-way clutch mechanism **15** are accommodated in the ratchet head **10**.

[0045] The power transmission mechanism **4** transmits the rotational force of the motor **2** to the socket **14**. The power transmission mechanism **4** includes a speed reduction mechanism **16** and an intermediate shaft **17**. The speed reduction mechanism **16** decelerates the rotation of the rotor **2**B (the motor **2**). The speed reduction mechanism **16** includes a planetary gear mechanism. The planetary gear mechanism of the speed reduction mechanism **16** is accommodated in the gear case **7**. The intermediate shaft **17** is accommodated in the extension case **9**.

[0046] The speed reduction mechanism **16** is disposed forward of the motor **2**. The speed reduction mechanism **16** is connected to the motor **2**. The speed reduction mechanism **16** includes a planetary gear mechanism. The planetary gear mechanism is rotated by the rotor **2**B. In the embodiment, the speed reduction mechanism **16** includes a front-stage planetary gear set and a rear-stage planetary gear set. That is, the speed reduction mechanism **16** has two-stages of planetary gears. The speed reduction mechanism **16** includes a pinion gear **16**A fixed to the front end portion of the rotor shaft **2**C of the motor **2**, a plurality of planetary gears **16**B disposed around the pinion gear **16**A, a carrier **16**C coupled to the planetary gears **16**B via pins, a plurality of planetary gears **16**D disposed around the front portion of the carrier **16**C, and a carrier **16**E coupled to the planetary gears **16**D via pins. The front portion of the carrier **16**C is a sun gear of the rear-stage planetary gear set. An internal gear **16**F is disposed around the pinion gear **16**A. The internal gear **16**F is provided on the inner circumferential face of the gear case **7**. The carrier **16**E is fixed to the rear end portion of the intermediate shaft **17**. The planetary gear mechanism of the speed reduction mechanism **16** is accommodated in the gear case **7**.

[0047] The intermediate shaft **17** is disposed forward of the speed reduction mechanism **16**. The intermediate shaft **17** is long in the front-rear direction. The intermediate shaft **17** rotates about the rotor axis AX.

[0048] The speed reduction mechanism **16** includes: a first bevel gear **18** that rotates about the rotor axis AX; and a second bevel gear **19** that is rotationally fixed to the socket **14**, meshes with the first bevel gear **18**, and rotates about the socket axis CX. The first bevel gear **18** and the second bevel gear **19** are accommodated in the ratchet head **10**.

[0049] The intermediate shaft **17** can connect the planetary gear mechanism and the first bevel gear **18** of the speed reduction mechanism **16**. The rear end portion of the intermediate shaft **17** is fixed to the carrier **16**E of the speed reduction mechanism **16**. The intermediate shaft **17** is rotationally fixed to the carrier **16**E. A pair of flats **66** is provided at the rear end portion of the intermediate shaft **17**. The pair of flats **66** provides a so-called width across flats at the rear end portion of the intermediate shaft **17**. The carrier **16**E has a tubular shape. When the rear end portion of the intermediate shaft **17** having the width across flats is inserted into the carrier **16**E, the intermediate shaft **17** is rotationally fixed to the carrier **16**E. The intermediate shaft **17** may be spline-coupled to the carrier **16**E or press-fitted into the carrier **16**E.

[0050] The rear portion of the intermediate shaft **17** is rotatably held by a bearing **20**. The bearing **20** is a ball bearing. A protrusion **17**T that contacts a front face of an inner ring of the bearing **20** is provided at the rear portion of the intermediate shaft **17**. The protrusion **17**T has an annular shape surrounding the rotor axis AX. The extension case **9** has a stepped portion that is in contact with the front face of the outer ring of the bearing **20**. The front end face of the gear case **7** is in contact with the rear face of the outer ring of the bearing **20**. The bearing **20** is positioned by the gear case **7**, the extension case **9**, and the intermediate shaft **17**.

[0051] The front end portion of the intermediate shaft **17** is fixed to the first bevel gear **18**. The front end portion of the intermediate shaft **17** has a tubular shape. The rear end portion of the first bevel gear **18** is inserted into the intermediate shaft **17**. The intermediate shaft **17** is rotationally fixed to the first bevel gear **18**. A pair of flats **67** is provided at the rear end portion of the first bevel gear **18**. The pair of flats **67** provides a so-called width across flats at the rear end portion of the first bevel gear **18**. The intermediate shaft **17** is rotationally fixed to the first bevel gear **18** by inserting the rear end portion of the first bevel gear **18** having the width across flats into the front end portion of the intermediate shaft **17**. The first bevel gear **18** may be spline-coupled to the intermediate shaft **17** or press-fitted to the intermediate shaft **17**.

[0052] The intermediate shaft **17** is accommodated in the extension case **9**. The lock nut **8** aligns the motor case **5** and the extension case **9** in the rotation direction. A screw thread is formed on the outer circumferential face of a front end portion of the motor case **5**. The inner circumferential face of the lock nut **8** has a screw groove **8**G, and is fixed to the outer circumferential face of the front end portion of the gear case **7**. The extension case **9** has a tubular shape. The inner circumferential face of the rear end portion of the extension case **9** has a screw groove **9**F, and is fixed to the outer circumferential face of the front end portion of the gear case **7**.

[0053] The rear end portion of the ratchet head **10** has a tubular shape. A screw thread **9**G is formed on the outer circumferential face of the front end portion of the extension case **9**. The inner circumferential face of the rear end portion of the ratchet head **10** has a screw groove **10**G, and is fixed to the outer circumferential face of the front end portion of the extension case **9**.
[0054] The first bevel gear **18** rotates about the rotor axis AX. The rear portion of the first bevel gear **18** is rotatably held by a bearing **21**. The bearing **21** is disposed inside the cylindrical rear end portion of the ratchet head **10**. An intermediate portion of the first bevel gear **18** is held by a needle bearing **22**. The needle bearing **22** is disposed inside a tubular rear end portion of the ratchet head **10**.
[0055] The second bevel gear **19** meshes with the first bevel gear **18**. The first bevel gear **18** has meshing teeth **18**G. The second bevel gear **19** has meshing teeth **19**G meshing with the meshing teeth **18**G of the first bevel gear **18**. The second bevel gear **19** is fixed to the lower portion of the socket **14**. The second bevel gear **19** rotates about the socket axis CX. The socket **14** rotates about the socket axis CX together with the second bevel gear **19**.
[0056] The motor housing **11** is disposed so as to cover the motor case **5**, the motor bracket **6**, the gear case **7**, the lock nut **8**, the extension case **9**, and the ratchet head **10**. The motor **2** is accommodated in the motor housing **11** via the motor case **5**. The motor housing **11** includes a handle portion **11**A gripped by a user of the electric screw tightening machine **1** and a battery holding portion **11**B disposed rearward of the handle portion **11**A.
[0057] The battery mounting portion **13** is disposed on lower portion of the battery holding portion **11**B. A battery pack **23** is attached to the battery mounting portion **13**. The battery pack **23** includes a rechargeable lithium-ion battery. The electric screw tightening machine **1** is a rechargeable electric screw tightening machine.
[0058] A paddle-shaped lever **24** is disposed on lower portion of the handle portion **11**A. When the lever **24** is operated by the user of the electric screw tightening machine **1**, the motor **2** is driven.
[0059] The operation member **12** is disposed on the upper portion of the handle portion **11**A. The operation member **12** is operated by the user of the electric screw tightening machine **1**. The operation member **12** is a dial rotatably supported by the handle portion **11**A. The operation member **12** is operated to operate the one-way clutch mechanism **15**. The one-way clutch mechanism **15** can switch between the free direction and the lock direction of the socket **14**. The operation member **12** is operated to switch between the free direction and the lock direction.
[0060] The operation member **12** and the one-way clutch mechanism **15** are coupled via a pair of wires **25**. The wires **25** include a wire **25**L disposed left of the ratchet head **10** and a wire **25**R disposed right of the ratchet head **10**. When the operation member **12** is rotated in one direction, the wire **25**L is pulled backward. When the operation member **12** is rotated in the other direction, the wire **25**R is pulled backward. When at least one of the wire **25**L and the wire **25**R is pulled backward, the one-way clutch mechanism **15** is operated.
[0061] The electric screw tightening machine **1** includes a wire dividing protrusion **61**, a wire introduction slit **62**, wire passage grooves **63**, wire guide protrusions **64**, and a plurality of protrusions **65** (**65**A, **65**B, **65**C).
[0062] The wire dividing protrusion **61** is disposed on the upper portion of the rear end portion of the motor case **5**. The wire **25** is divided into a wire **25**L and a wire **25**R by the wire dividing protrusion **61** at the rear end portion of the motor case **5**.
[0063] The wire introduction slit **62** is disposed on upper portion of the rear end portion of the motor case **5**. The wire dividing protrusion **61** is disposed in the wire introduction slit **62**. An annular rib **62**A is provided at the rear end portion of the motor case **5**. The wire introduction slit **62** is provided at upper portion of the rib **62**A.
[0064] The wire passage grooves **63** allow the wire **25** to pass therethrough. The wire passage grooves **63** are provided forward of the wire dividing protrusion **61**. The wire passage grooves **63** are provided at the upper face of the motor case **5** so as to extend in the front-rear direction. The

pair of wire passage grooves **63** is provided in the left-right direction. The wire **25**L passes through the left-side wire passage groove **63**. The wire **25**R passes through the right-side wire passage groove **63**.
[0065] The wire guide protrusions **64** guide the wire **25** forward. The wire guide protrusions **64** are disposed at upper portion of the front end portion of the motor case **5**. The wire guide protrusions **64** are provided forward of the respective wire passage grooves **63**. The pair of wire guide protrusions **64** is provided in the left-right direction. The wire **25**L is guided by the left-side wire guide protrusion **64**. The wire **25**R is guided by the right-side wire guide protrusion **64**.
[0066] The protrusion **65** is provided on each of the left side face and the right side face of each of the extension case **9** and the ratchet head **10**. The protrusions **65** guide the wire **25**. The protrusions **65** includes: a protrusion **65**A, a protrusion **65**B, and a protrusion **65**C each protruding leftward from the left side faces of the extension case **9** and the ratchet head **10**; and a protrusion **65**A, a protrusion **65**B, and a protrusion **65**C each protruding rightward from the right side faces of the extension case **9** and the ratchet head **10**. The protrusion **65**A, the protrusion **65**B, and the protrusion **65**C are disposed in the front-rear direction. The protrusion **65**A is disposed at the rear portion of the extension case **9**. The protrusion **65**B is disposed forward and downward of the protrusion **65**A at the rear portion of the extension case **9**. The protrusion **65**C is disposed at the rear portion of the ratchet head **10**.
[0067] A wire hole **68** is provided at each of the left side face and the right side face of the ratchet head **10**. The wire **25** having passed through the outside of the extension case **9** is inserted into the inside of the ratchet head **10** through the wire hole **68**.
[0068] The socket **14** extends in the vertical direction. The socket **14** has the socket axis CX that intersects with the rotor axis AX. Rotation of the motor **2** is always transmitted to the socket **14** via the speed reduction mechanism **16** during rotation of the motor **2**. The socket **14** is rotated by the planetary gear mechanism of the speed reduction mechanism **16**. A socket adapter **26** as a distal end tool is attached to the socket **14**. The upper portion of the socket adapter **26** is inserted into the socket **14**. A replacement socket (not illustrated) is attached to the lower portion of the socket adapter **26** protruding downward from the socket **14**. The fastening work is performed by rotating the socket **14** in a state where the screw, the bolt, or the nut is inserted into the replacement socket.
[0069] The sensor **27** detects an operation state of the operation member **12**. The sensor **27** detects the rotation direction of the operation member **12**. By detecting the operation state of the operation member **12**, the sensor **27** detects that the wire **25**L or the wire **25**R is pulled backward.
[0070] As illustrated in FIG. **3**, the sensor **27** is mounted on a sensor substrate **27**P. The sensor substrate **27**P is supported by the rear portion of the motor case **5**. The sensor substrate **27**P is disposed downward of the operation member **12**. A magnet **28** is fixed to the operation member **12**. The sensor **27** is disposed downward of the magnet **28**. The sensor **27** can face the magnet **28**. When the operation member **12** is operated, the position of the magnet **28** changes. The sensor **27** is a Hall IC (magnetic sensor). The sensor **27** can detect the operation state of the operation member **12** by detecting a change in the magnetic force of the magnet **28**. A detection signal of the sensor **27** is transmitted to the controller **50**. The controller **50** controls the rotation direction of the motor **2** based on the detection signal from the sensor **27**.
[0071] The controller **50** controls at least the motor **2**. The detection signal of the sensor **27** is input to the controller **50**. The controller **50** controls the rotation direction of the motor **2** based on the detection signal of the sensor **27**.
[0072] The controller **50** includes a circuit board **50**A on which a plurality of electronic components is mounted. The controller **50** is disposed inside the battery holding portion **11**B of the motor housing **11**. Examples of the electronic components mounted on the circuit board **50**A include a microcomputer **50**B, a switching element **50**C such as an MOSDET, and a capacitor **50**D. Six switching elements **50**C are provided. Note that the electronic components mounted on the circuit board **50**A may include a transistor and a resistor.

Output Unit

[0073] FIG. **5** is a perspective view from the front illustrating the output unit **3** according to the embodiment. FIG. **6** is a longitudinal sectional view illustrating the output unit **3** according to the embodiment. FIG. **7** is a transverse cross-sectional view illustrating the output unit **3** according to the embodiment. FIG. **8** is an exploded perspective view from above illustrating the output unit **3** according to the embodiment. FIG. **9** is an exploded perspective view from below illustrating the output unit **3** according to the embodiment. FIG. **10** is a perspective view of an extracted main part of the output unit **3** according to the embodiment. FIG. **11** is a view of the ratchet head **10** according to the embodiment when viewed from below.

[0074] The second bevel gear **19** is disposed around the lower portion of the socket **14**. The second bevel gear **19** and the socket **14** are fixed. The meshing teeth **18**G of the first bevel gear **18** meshes with the meshing teeth **19**G of the second bevel gear **19**. When the first bevel gear **18** rotates, the second bevel gear **19** rotates. When the second bevel gear **19** rotates, the socket **14** rotates together with the second bevel gear **19**. A gear **14**G is provided on an outer face of the socket **14**.

[0075] The one-way clutch mechanism **15** is disposed at least partially around the socket **14**. During rotation of the motor **2**, the one-way clutch mechanism **15** allows the socket **14** to rotate only in one direction by the rotation of the motor **2**. The socket **14** rotates in the free direction by the rotational force of the motor **2**. During rotation of the motor **2**, the socket **14** rotates at 800 rpm or more. The socket **14** may rotate at 820 rpm or more. The socket **14** may rotate at 840 rpm or more. The socket **14** may rotate at 850 rpm or more. The socket **14** may rotate at 860 rpm or more. The socket **14** may rotate at 880 rpm or more. The socket **14** may rotate at 900 rpm or more. The socket **14** may rotate at 1000 rpm or more. The rotation speed of the socket **14** is the maximum rotation speed of the socket **14** by the motor **2**.

[0076] The maximum rotation speed of the socket **14** may be 500 rpm or more. While the maximum rotation speed of the socket **14** is set to 500 rpm or more, the maximum tightening torque by the socket **14** may be set to 50 N.Math.m or more. The maximum tightening torque by the socket **14** may be set to 110 N.Math.m or more.

[0077] When the motor **2** is stopped, the electric screw tightening machine **1** functions as a manual ratchet wrench. When the electric screw tightening machine **1** is used as a manual ratchet wrench, the user grips the handle portion **11**A of the motor housing **11** and operates the motor housing **11** so that the motor housing **11** rotates (turns) around the socket **14**. The rotational force input from the user to the motor housing **11** is transmitted to the socket **14**. During stop of the motor **2**, the one-way clutch mechanism **15** transmits the rotational force input from the user to the motor housing **11** to the socket **14** only in one direction.

[0078] The one-way clutch mechanism **15** includes a cam plate **30**, a first ratchet pawl **31**, a second ratchet pawl **32**, a first spring **33**, a second spring **34**, a positioning ball **35**, a spring **36**, a washer **37**, a washer **38**, a retainer ring **39**, and a screw **40**. At least the first ratchet pawl **31** and the second ratchet pawl **32** of the one-way clutch mechanism **15** are ratchet portions connectable to the socket **14**.

[0079] The cam plate **30** is disposed at least partially around the socket **14**. The cam plate **30** is rotatable about the socket axis CX with respect to the ratchet head **10**. The cam plate **30** is disposed upward of the second bevel gear **19**. The cam plate **30** includes a base plate **30**A disposed partially around the socket **14**, a cam projection **30**B projecting upward from the upper face of the base plate **30**A, a first recess **30**D and a second recess **30**C provided at the upper face of the base plate **30**A, a screw hole **30**F, a pin release recess **30**L, a pin release recess **30**R, and a wire support projection **30**S.

[0080] The cam plate **30** is rotatable about the socket axis CX by operating the operation member **12**. The cam plate **30** is rotatable in each of a forward rotation direction La and a reverse rotation direction Ra illustrated in FIG. **7**.

[0081] The socket **14**, the first ratchet pawl **31**, the second ratchet pawl **32**, the first spring **33**, and the second spring **34** are disposed in a first internal space **10**A of the ratchet head **10**. Each of the first ratchet pawl **31** and the second ratchet pawl **32** is movable inside the first internal space **10**A. The cam projection **30**B is disposed between the first ratchet pawl **31** and the second ratchet pawl **32** in the circumferential direction of the socket axis CX. The first ratchet pawl **31** is disposed on the forward rotation direction La side (left side) of the cam projection **30**B. The second ratchet pawl **32** is disposed on the reverse rotation direction Ra side (right side) of the cam projection **30**B.

[0082] The first ratchet pawl **31** and the second ratchet pawl **32** are relatively movable. The first ratchet pawl **31** has a gear **31**G that can mesh with the gear **14**G of the socket **14**. The second ratchet pawl **32** has a gear **32**G that can mesh with the gear **14**G of the socket **14**.

[0083] The first ratchet pawl **31** is movably supported by the upper face of the base plate **30**A of the cam plate **30**. The first ratchet pawl **31** and the cam plate **30** are relatively movable. The first ratchet pawl **31** is movable in the first internal space **10**A while being guided by a first guide face **10**C which is part of the inner face of the first internal space **10**A.

[0084] The second ratchet pawl **32** is movably supported by the upper face of the base plate **30**A of the cam plate **30**. The second ratchet pawl **32** and the cam plate **30** are relatively movable. The second ratchet pawl **32** is movable in the first internal space **10**A while being guided by a second guide face **10**D which is part of the inner face of the first internal space **10**A.

[0085] One end of the first spring **33** is attached to the inner face of the first internal space **10**A. The other end of the first spring **33** is attached to the first ratchet pawl **31**. The first spring **33** generates an elastic force so as to press the first ratchet pawl **31** against the cam projection **30**B.

[0086] One end of the second spring **34** is attached to the inner face of the first internal space **10**A. The other end of the second spring **34** is attached to the second ratchet pawl **32**. The second spring **34** generates an elastic force so as to press the second ratchet pawl **32** against the cam projection **30**B.

[0087] The base plate **30**A of the cam plate **30** has an arc shape disposed partially around the socket **14**. The cam projection **30**B protrudes upward from the upper face of the base plate **30**A. The cam projection **30**B has an arc shape disposed partially around the socket **14**. The first recess **30**D and a second recess **30**C are provided on the upper face of the base plate **30**A. The second recess **30**C is provided left of the first recess **30**D. The screw hole **30**F is provided in the base plate **30**A. The screw hole **30**F is provided left of the second recess **30**C. The pin release recess **30**L is provided so as to be recessed in the reverse rotation direction Ra from the left end portion of the base plate **30**A. The pin release recess **30**R is provided so as to be recessed in the forward rotation direction La from the right end portion of the base plate **30**A. The wire support projection **30**S protrudes downward from the inner edge of the lower face of the base plate **30**A. The wire support projection **30**S has an arc shape disposed partially around the socket **14**.

[0088] The washer **37** is disposed between the cam plate **30** and the second bevel gear **19**. The washer **37** prevents contact between the cam plate **30** and the second bevel gear **19**. The washer **37** does not rotate with respect to the ratchet head **10**. The washer **37** has an arc shape disposed partially around the socket **14**. The washer **37** has a screw release hole **37**F, a pin hole **37**L, and a pin hole **37**R. The screw release hole **37**F is provided at the center of the washer **37** in the circumferential direction of the socket axis CX. The screw release hole **37**F is a long hole that is long in the left-right direction. The pin hole **37**L is provided at the left end portion of the washer **37**. The pin hole **37**R is provided at the right end portion of the washer **37**.

[0089] The washer **38** is disposed below the second bevel gear **19**. The washer **38** supports the second bevel gear **19** from below. The retainer ring **39** supports the washer **38** from below. The retainer ring **39** is inserted into a notch provided in the ratchet head **10**.

[0090] A part of the wire **25** and the cam plate **30** are fixed by the screw **40**. The front end portion of the wire **25** is disposed so as to wind a part of the wire support projection **30**S of the cam plate **30**. The front end portion of the wire **25** is hooked on the wire support projection **30**S of the cam plate **30**. The screw portion of the screw **40** is inserted into the screw hole **30**F of the cam plate **30**.

The screw portion of the screw **40** is coupled to the screw hole **30**F. By coupling the screw portion of the screw **40** and the screw hole **30**F, the screw **40** is fixed to the cam plate **30**. As illustrated in FIG. **6**, the front end portion of the wire **25** is sandwiched between the head portion of the screw **40** and the base plate **30**A. As a result, a part of the wire **25** and the cam plate **30** are fixed by the screw **40**.

[0091] The head portion of the screw **40** is disposed in the screw release hole **37**F of the washer **37**. When the cam plate **30** rotates about the socket axis CX, the screw **40** turns around the socket axis CX. Since the head portion of the screw **40** is disposed in the screw release hole **37**F which is a long hole, the rotation of the cam plate **30** is not hindered by the washer **37**.

[0092] As illustrated in FIG. **7**, the output unit **3** includes a pin **70**L and a pin **70**R. The pin **70**L is disposed left rear of the socket **14**. The pin **70**R is disposed right rear of the socket **14**. The washer **37** has the pin hole **37**L into which the pin **70**L is inserted and the pin hole **37**R into which the pin **70**R is inserted. The cam plate **30** has the pin release recess **30**L in which the pin **70**L is disposed and the pin release recess **30**R in which the pin **70**R is disposed. The ceiling face of the first internal space **10**A of the ratchet head **10** is provided with a pin recess **10**L in which the upper end portion of the pin **70**L is disposed and a pin recess **10**R in which the upper end portion of the pin **70**R is disposed. The washer **37** is positioned on the ratchet head **10** by the pin **70**L and the pin **70**R. Since the pin **70**L is disposed at the pin release recess **30**L and the pin **70**R is disposed at the pin release recess **30**R, the rotation of the cam plate **30** is not hindered by the pin **70**L and the pin **70**R.

[0093] When the operation member **12** is operated by the user and the wire **25** is pulled, the cam plate **30** rotates. When the wire **25**L is pulled backward, the cam plate **30** rotates in the forward rotation direction La. When the wire **25**R is pulled backward, the cam plate **30** rotates in the reverse rotation direction Ra.

[0094] When the wire **25**L is pulled backward and the cam plate **30** rotates in the forward rotation direction La, the first ratchet pawl **31** is pushed in the forward rotation direction La by the cam projection **30**B. The first ratchet pawl **31** moves in the forward rotation direction La while being guided by the first guide face **10**C against the elastic force of the first spring **33**. When the first ratchet pawl **31** moves in the forward rotation direction La, the first ratchet pawl **31** moves radially outward of the socket axis CX. That is, when the first ratchet pawl **31** moves in the forward rotation direction La, the first ratchet pawl **31** is away from the socket **14**.

[0095] When the wire **25**R is pulled backward and the cam plate **30** rotates in the reverse rotation direction Ra, the second ratchet pawl **32** is pushed in the reverse rotation direction Ra by the cam projection **30**B. The second ratchet pawl **32** moves in the reverse rotation direction Ra while being guided by the second guide face **10**D against the elastic force of the second spring **34**. As illustrated in FIG. **7**, when the second ratchet pawl **32** moves in the reverse rotation direction Ra, the second ratchet pawl **32** moves radially outward of the socket axis CX. That is, when the second ratchet pawl **32** moves in the reverse rotation direction Ra, the second ratchet pawl **32** is away from the socket **14**.

[0096] The positioning ball **35** is disposed in a second internal space **10**B of the ratchet head **10**. The positioning ball **35** is disposed in one of the first recess **30**D and the second recess **30**C. The spring **36** biases the positioning ball **35** downward. The upper end of the spring **36** is attached to the ceiling face of the second internal space **10**B. In the embodiment, the ceiling face of the second internal space **10**B has a recess **10**H in which the upper end of the spring **36** is accommodated. The upper end of the spring **36** is positioned in the ratchet head **10** by being disposed in the recess **10**H. The lower end of the spring **36** is attached to the positioning ball **35**.

[0097] The positioning ball **35** positions the cam plate **30**. When the cam plate **30** moves in the forward rotation direction La so that the first ratchet pawl **31** is away from the socket **14**, the positioning ball **35** is disposed in the first recess **30**D. When the cam plate **30** rotates in the reverse rotation direction Ra so that the second ratchet pawl **32** is away from the socket **14**, the positioning ball **35** is disposed in the second recess **30**C.

[0098] In a state where the cam plate **30** moves in the forward rotation direction La so that the first ratchet pawl **31** is away from the socket **14**, the gear **32**G of the second ratchet pawl **32** and the gear **14**G of the socket **14** mesh with each other. The socket **14** cannot rotate in the forward rotation direction La (counterclockwise) but can rotate in the reverse rotation direction Ra (clockwise). That is, when the cam plate **30** rotates in the forward rotation direction La and the first ratchet pawl **31** is away from the socket **14**, the reverse rotation direction Ra (clockwise) is the free direction of the socket **14**, and the forward rotation direction La (counterclockwise) is the lock direction of the socket **14**.

[0099] As illustrated in FIG. **7**, in a state where the cam plate **30** moves in the reverse rotation direction Ra so that the second ratchet pawl **32** is away from the socket **14**, the gear **31**G of the first ratchet pawl **31** and the gear **14**G of the socket **14** mesh with each other. The socket **14** cannot rotate in the reverse rotation direction Ra (clockwise) but can rotate in the forward rotation direction La (counterclockwise). That is, when the cam plate **30** rotates in the reverse rotation direction Ra and the second ratchet pawl **32** is away from the socket **14**, the forward rotation direction La (counterclockwise) is the free direction of the socket **14**, and the reverse rotation direction Ra (clockwise) is the lock direction of the socket **14**.

[0100] When the socket **14** is rotated in the forward rotation direction La (counterclockwise) by the rotational force of the motor **2**, the user operates the operation member **12** so that forward rotation direction La (counterclockwise) is the free direction of the socket **14**. That is, the user operates the operation member **12** so that the second ratchet pawl **32** is away from the socket **14**. The operation state of the operation member **12** is detected by the sensor **27**. When determining that the forward rotation direction La (counterclockwise) of the socket **14** is the free direction of the socket **14** based on the detection data of the sensor **27**, the controller **50** controls the rotation direction of the motor **2** so that the motor **2** rotates the socket **14** in the forward rotation direction La (counterclockwise).

[0101] When the socket **14** is rotated in the reverse rotation direction Ra (clockwise) by the rotational force of the motor **2**, the user operates the operation member **12** so that reverse rotation direction Ra (clockwise) is the free direction of the socket **14**. That is, the user operates the operation member **12** so that the first ratchet pawl **31** is away from the socket **14**. The operation state of the operation member **12** is detected by the sensor **27**. When determining that the reverse rotation direction Ra (clockwise) of the socket **14** is the free direction of the socket **14** based on the detection signal of the sensor **27**, the controller **50** controls the rotation direction of the motor **2** so that the motor **2** rotates the socket **14** in the reverse rotation direction Ra (clockwise).

[0102] For example, when the screw tightening work is performed, the operation member **12** is operated so that the socket **14** rotates in the reverse rotation direction Ra (clockwise), that is, the reverse rotation direction Ra (clockwise) is the free direction. The motor **2** generates a rotational force about the rotor axis AX so that the socket **14** rotates in the reverse rotation direction Ra (clockwise).

[0103] After tightening the screw owing to the rotational force of the motor **2**, the user may additionally tighten the screw. In the embodiment, the electric screw tightening machine **1** functions as a manual ratchet wrench during the stop of the motor **2**. After tightening the screw by the rotational force of the motor **2**, the user operates the electric screw tightening machine **1** that functions as a manual ratchet wrench to additionally tighten the screw while the motor **2** is stopped.

[0104] When the screw is fastened by the rotational force of the motor **2**, the operation member **12** is operated so that the reverse rotation direction Ra (clockwise) of the socket **14** is the free direction of the socket **14**. When the electric screw tightening machine **1** is used as a manual ratchet wrench to additionally tighten the screw, the user grips the handle portion **11**A of the motor housing **11** and operates the motor housing **11** so that the motor housing **11** rotates (turns) around the socket **14**. Since the forward rotation direction La (counterclockwise) of the socket **14** is the lock direction of the socket **14**, the user can additionally tighten the screw by rotating (turning) the motor housing **11** around the socket **14** in the reverse rotation direction Ra (clockwise). The user can rotate (turn) the motor housing **11** around the socket **14** in the forward rotation direction La (counterclockwise) while maintaining the position of the socket **14** in the rotation direction. By repeating the operation of turning the motor housing **11** in the reverse rotation direction Ra and the operation of turning the motor housing **11** in the forward rotation direction La, the user can use the electric screw tightening machine **1** in a state where the motor **2** is stopped as a manual ratchet wrench.

Attachment and Detachment of Intermediate Shaft

[0105] FIG. **12** is a side view illustrating the electric screw tightening machine **1**B according to the embodiment. The intermediate shaft **17** is attachable to and detachable from the planetary gear mechanism and the first bevel gear **18** of the speed reduction mechanism **16**. When the intermediate shaft **17** is attached to each of the planetary gear mechanism and the first bevel gear **18** of the speed reduction mechanism **16**, the rotation of the planetary gear mechanism of the speed reduction mechanism **16** can be transmitted to the first bevel gear **18** via the intermediate shaft **17**. When the intermediate shaft **17** is detached from each of the planetary gear mechanism and the first bevel gear **18** of the speed reduction mechanism **16**, the rotation of the planetary gear mechanism of the speed reduction mechanism **16** can be transmitted to the first bevel gear **18** without via the intermediate shaft **17**.

[0106] After the intermediate shaft **17** is removed, the speed reduction mechanism **16** and the first bevel gear **18** are directly coupled. The extension case **9** is attachable to and detachable from the gear case **7** and the ratchet head **10**. After the extension case **9** is removed, the gear case **7** and the ratchet head **10** are directly coupled. As illustrated in FIG. **12**, the intermediate shaft **17** and the extension case **9** are removed, whereby the dimension of the electric screw tightening machine **1** in the front-rear direction decreases.

Effects

[0107] As described above, in the embodiment, the electric screw tightening machine **1** includes: the motor **2** including the stator **2**A and the rotor **2**B that is rotatable with respect to the stator **2**A and has the rotor axis AX extending in the front-rear direction; the motor housing **11** that accommodates the motor **2**; the speed reduction mechanism **16** that decelerates the rotation of the rotor **2**B (the motor **2**); the socket **14** to which rotation of the motor **2** is always transmitted via the speed reduction mechanism **16** during rotation of the motor **2** and that has the socket axis CX intersecting with the rotor axis AX; and the one-way clutch mechanism **15** that is disposed at least partially around the socket **14**, and is configured to allow the socket **14** to rotate due to the rotation of the motor **2** only in one direction during rotation of the motor **2** and transmit the rotation input to the motor housing **11** only in one direction to the socket **14** during stop of the motor **2**.

[0108] In the above configuration, for example, when the screw is tightened, the socket **14** is rotated by the rotational force of the motor **2**. Since the motor **2** rotates at a high speed, the socket **14** also rotates at a high speed. Since the socket **14** rotates at a high speed, the screw is tightened in a short time. During stop of the motor **2**, the electric screw tightening machine **1** can function as a manual ratchet wrench. Therefore, after tightening the screw by the rotational force of the motor **2**, the user of the electric screw tightening machine **1** can additionally tighten the screw by using the electric screw tightening machine **1** with the motor **2** stopped as a manual ratchet wrench. That is, after tightening the screw at a high speed to some extent by the rotational force of the motor **2**, the user can additionally tighten the screw by using the electric screw tightening machine **1** with the motor **2** stopped as a manual ratchet wrench. Therefore, a decrease in work efficiency is suppressed.

[0109] In the embodiment, during rotation of the motor **2**, the socket **14** rotates at 800 rpm or more.

[0110] In the above configuration, since the socket **14** rotates at a high speed, the tightening work is completed in a short time.

[0111] In the embodiment, the electric screw tightening machine **1** functions as a manual ratchet wrench during the stop of the motor.

[0112] In the above configuration, after tightening the screw at a high speed by the rotational force of the motor **2**, the user can additionally tighten the screw by the electric screw tightening machine **1** with the motor **2** stopped.
[0113] In the embodiment, the speed reduction mechanism **16** includes: the first bevel gear **18** that rotates about the rotor axis AX; and the second bevel gear **19** that is rotationally fixed to the socket **14**, meshes with the first bevel gear **18**, and rotates about the socket axis CX.
[0114] In the above configuration, since the rotor axis AX and the socket axis CX are orthogonal to each other, the electric screw tightening machine **1** can be used as a manual ratchet wrench. The rotational force of the motor **2** is transmitted to the socket **14** via the first bevel gear **18** and the second bevel gear **19**.
[0115] In the embodiment, the electric screw tightening machine **1** includes the intermediate shaft **17** configured to couple the planetary gear mechanism and the first bevel gear **18** of the speed reduction mechanism **16**. The electric screw tightening machine **1** is configured such that the rotation of the speed reduction mechanism **16** is transmitted to the first bevel gear **18** via the intermediate shaft **17**, and the rotation of the speed reduction mechanism **16** is transmitted to the first bevel gear **18** without via the intermediate shaft **17**.
[0116] In the above configuration, the dimension of the electric screw tightening machine **1** in the front-rear direction can be adjusted in any dimension.
[0117] In the embodiment, the one-way clutch mechanism **15** allows the rotation of the socket **14** in the free direction and blocks the socket **14** from rotating in the lock direction. The one-way clutch mechanism is configured to switch between the free direction and the lock direction. The electric screw tightening machine **1** includes the operation member **12** that is operated so as to switch between a free direction and a lock direction.
[0118] In the above configuration, the user can perform, for example, both the screw tightening work and the screw loosening work using the electric screw tightening machine **1**.
[0119] In the embodiment, the electric screw tightening machine **1** includes: the sensor **27** that detects an operation state of the operation member **12**; and the controller **50** to which a detection signal of the sensor **27** is input. The controller **50** controls the rotation direction of the motor **2** such that the motor **2** rotates the socket **14** in the forward rotation direction when determining that the forward rotation direction of the socket **14** is the free direction of the socket **14** based on the detection signal of the sensor **27**, and the motor **2** rotates the socket **14** in the reverse rotation direction when determining that the reverse rotation direction of the socket **14** is the free direction of the socket **14**.
[0120] In the above configuration, the controller **50** can control the rotation direction of the motor **2** so that the rotational force of the motor **2** is not transmitted in the lock direction of the socket **14** but is transmitted in the free direction. After tightening the screw by the rotational force of the motor **2**, the user can additionally tighten the screw with the electric screw tightening machine **1** functioning as a manual ratchet wrench without operating the operation member **12**.
[0121] In the embodiment, the one-way clutch mechanism **15** includes: the cam plate **30** that is disposed at least partially around the socket **14** and rotates about the socket axis CX by the operation of the operation member **12**; the first ratchet pawl **31** that is movably supported by the upper face of the cam plate **30** and meshes with the gear of the socket **14**; the second ratchet pawl **32** that is movably supported by the upper face of the cam plate **30** and meshes with the gear of the socket **14**; the first spring **33** that generates an elastic force so as to press the first ratchet pawl **31** against the cam projection **30**B provided on the upper face of the cam plate **30**; and the second spring **34** that generates an elastic force so as to press the second ratchet pawl **32** against the cam projection **30**B. When the cam plate **30** rotates in the forward rotation direction, the first ratchet pawl **31** is away from the socket **14**, the reverse rotation direction is the free direction of the socket **14**, and the forward rotation direction is the lock direction of the socket **14**. When the cam plate **30** rotates in the reverse rotation direction, the second ratchet pawl **32** is away from the socket **14**, the

forward rotation direction is the free direction of the socket **14**, and the reverse rotation direction is the lock direction of the socket **14**.
[0122] In the above configuration, the free direction and the lock direction of the socket **14** are switched only by operating the operation member **12**.
[0123] In the embodiment, the electric screw tightening machine **1** includes the positioning ball **35** that is to be disposed in the first recess **30**D provided in the cam plate **30** when the cam plate **30** rotates in the forward rotation direction, and is to be disposed in the second recess **30**C provided in the cam plate **30** when the cam plate **30** rotates in the reverse rotation direction.
[0124] In the above configuration, the cam plate **30** is positioned by the positioning ball **35**.
[0125] Although the invention has been described with respect to specific embodiments for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art that fairly fall within the basic teaching herein set forth.

Claims

1. An electric screw tightening machine comprising: a motor including a stator and a rotor, the rotor being rotatable with respect to the stator and having a rotor axis extending a front-rear direction; a motor housing that accommodates the motor; a speed reduction mechanism that decelerates rotation of the motor; a socket to which rotation of the motor is always transmitted via the speed reduction mechanism during rotation of the motor, the socket having a socket axis that intersects with the rotor axis; and a one-way clutch mechanism that is disposed at least partially around the socket and is configured to allow the socket to rotate due to the rotation of the motor only in one direction during the rotation of the motor, and transmit rotation input to the motor housing only in one direction to the socket during stop of the motor.
2. The electric screw tightening machine according to claim 1, wherein the socket rotates at 800 rpm or more during the rotation of the motor.
3. The electric screw tightening machine according to claim 2, wherein the electric screw tightening machine functions as a manual ratchet wrench during the stop of the motor.
4. The electric screw tightening machine according to claim 1, wherein the speed reduction mechanism includes a first bevel gear that rotates about the rotor axis, and a second bevel gear that is rotationally fixed to the socket, meshes with the first bevel gear, and rotates about the socket axis.
5. The electric screw tightening machine according to claim 4, further comprising an intermediate shaft configured to connect a planetary gear mechanism and the first bevel gear of the speed reduction mechanism, wherein the electric screw tightening machine is configured such that rotation of the planetary gear mechanism of the speed reduction mechanism is transmitted to the first bevel gear via the intermediate shaft, and rotation of the planetary gear mechanism of the speed reduction mechanism is transmitted to the first bevel gear without via the intermediate shaft.
6. The electric screw tightening machine according to claim 1, wherein the one-way clutch mechanism allows the socket to rotate in a free direction and blocks the socket from rotating in a lock direction, and the one-way clutch mechanism is configured to switch between the free direction and the lock direction, and the electric screw tightening machine comprises an operation member that is operated to switch between the free direction and the lock direction.
7. The electric screw tightening machine according to claim 6, further comprising: a sensor that detects an operation state of the operation member; and a controller to which a detection signal of the sensor is input, wherein the controller controls a rotation direction of the motor such that the motor rotates the socket in a forward rotation direction when determining that the forward rotation direction of the socket is the free direction of the socket based on the detection signal of the sensor, and the motor rotates the socket in a reverse rotation direction when determining that the reverse

rotation direction of the socket is the free direction of the socket based on the detection signal of the sensor.

8. The electric screw tightening machine according to claim 7, wherein the one-way clutch mechanism includes a cam plate that is disposed at least partially around the socket and rotates about the socket axis in response to an operation of the operation member, a first ratchet pawl that is movably supported by an upper face of the cam plate and meshes with a gear of the socket, a second ratchet pawl that is movably supported by an upper face of the cam plate and meshes with a gear of the socket, a first spring that generates an elastic force so as to press the first ratchet pawl against a cam projection provided on the upper face of the cam plate, and a second spring that generates an elastic force so as to press the second ratchet pawl against the cam projection, when the cam plate rotates in the forward rotation direction, the first ratchet pawl is away from the socket, the reverse rotation direction is the free direction of the socket, and the forward rotation direction is the lock direction of the socket, and when the cam plate rotates in the reverse rotation direction, the second ratchet pawl is away from the socket, the forward rotation direction is the free direction of the socket, and the reverse rotation direction is the lock direction of the socket.

9. The electric screw tightening machine according to claim 8, further comprising a positioning ball that is to be disposed in a first recess provided in the cam plate when the cam plate rotates in the forward rotation direction, and is to be disposed in a second recess provided in the cam plate when the cam plate rotates in the reverse rotation direction.

10. The electric screw tightening machine according to claim 1, wherein the socket has a maximum rotation speed of 850 rpm or more.

11. The electric screw tightening machine according to claim 1, wherein the socket has a maximum rotation speed of 500 rpm or more and has a maximum tightening torque of 50 N.Math.m or more.

12. The electric screw tightening machine according to claim 1, wherein the socket has a maximum tightening torque of 110 N.Math.m or more.

13. An angle ratchet wrench comprising: a brushless motor including a stator and a rotor rotatable radially inside the stator, the brushless motor extending in a front-rear direction; a planetary gear rotated by the rotor; a socket that is rotated by the planetary gear and extends in an up-down direction; and a ratchet portion connectable to the socket, wherein the socket has a maximum rotation speed of 850 rpm or more.

14. An angle ratchet wrench comprising: a brushless motor including a stator and a rotor rotatable radially inside the stator, the brushless motor extending in a front-rear direction; a planetary gear rotated by the rotor; a socket that is rotated by the planetary gear and extends in an up-down direction; and a ratchet portion connectable to the socket, wherein the socket has a maximum rotation speed of 500 rpm or more and has a maximum tightening torque of 50 N.Math.m or more.

15. An angle ratchet wrench comprising: a brushless motor including a stator and a rotor rotatable radially inside the stator, the brushless motor extending in a front-rear direction; a planetary gear rotated by the rotor; a socket that is rotated by the planetary gear and extends in an up-down direction; and a ratchet portion connectable to the socket, wherein the socket has a maximum tightening torque of 110 N.Math.m or more.